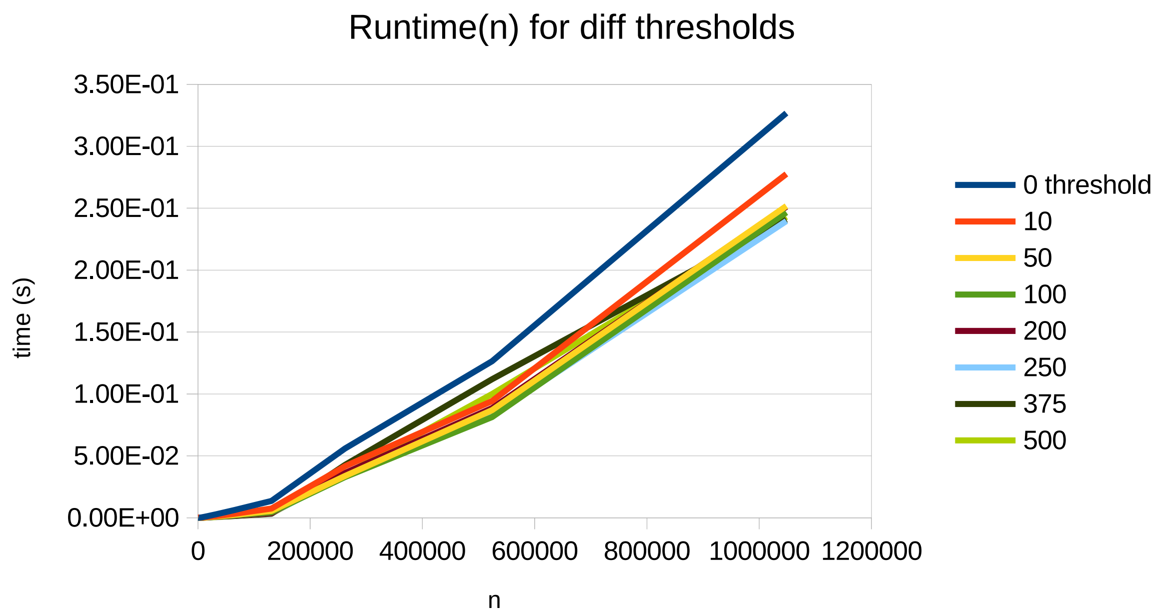
Jon Pulsipher - 6012 - A4, MergeSort/QuickSort Analysis

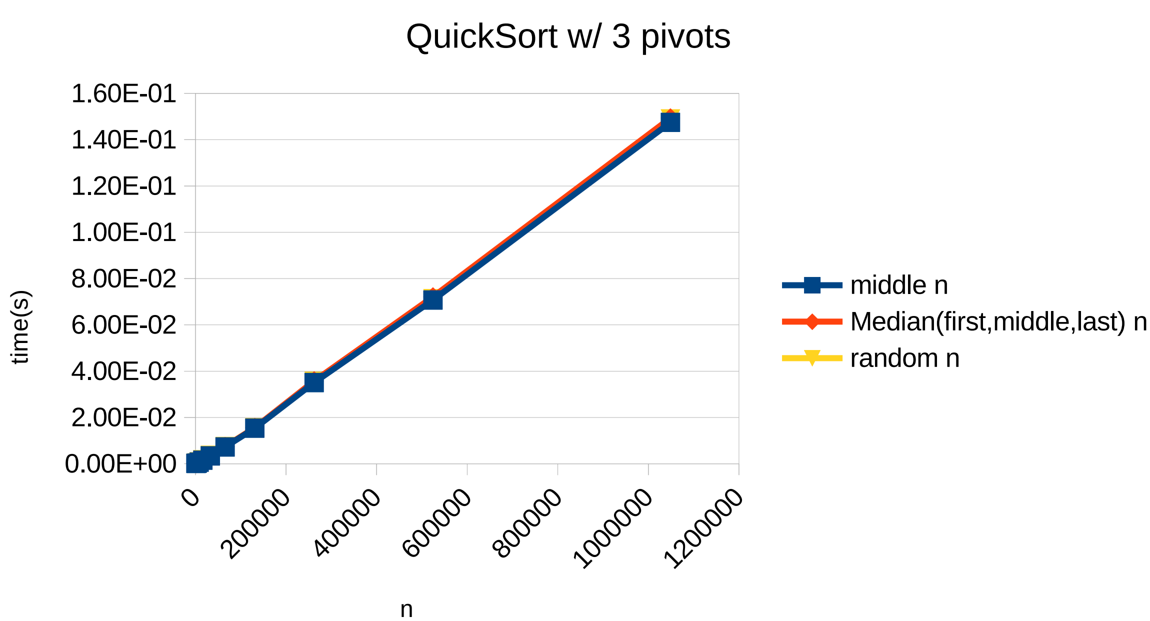
1) Mergesort Threshold Experiment

A threshold of **250** had the best performance out of those sampled. I didn't dig deeper than that or explore length-based thresholds.



2) Pivot Method Experiment

Finding the best Pivot method was not as straight-forward. 'Middle', 'random', and 'median' all performed about the same on a shuffled sample. Middle slightly edged out the other two pivot selection methods.



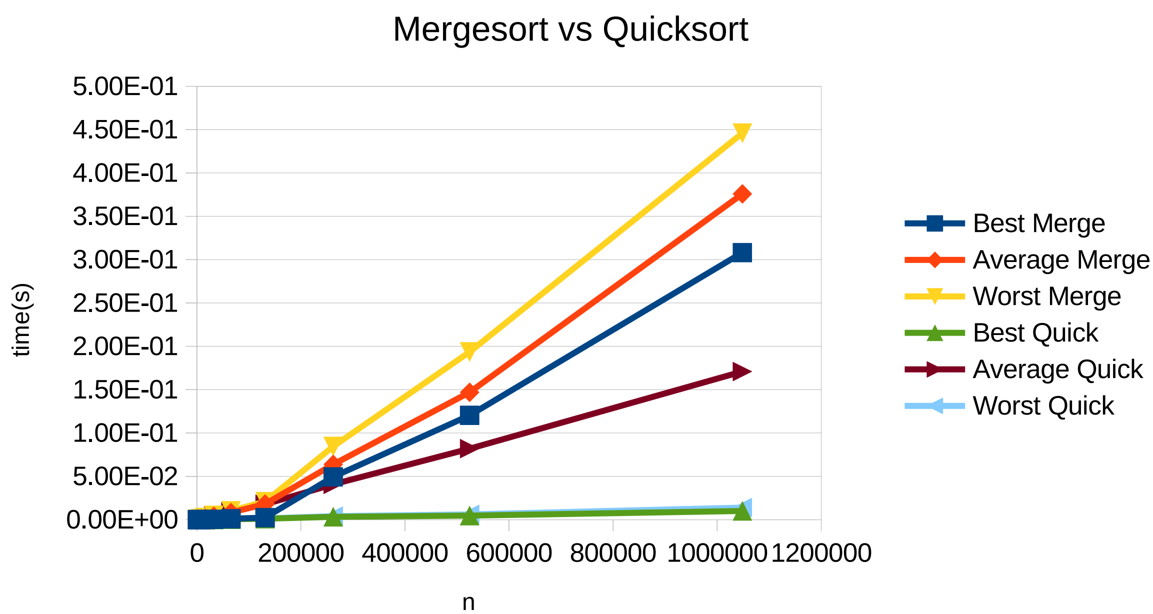
This does make sense – on a randomly shuffled data-set there you should expect 'random' and 'middle' to give similarly random pivots, and any small edge 'median' gets from sampling three indices may be outweighed by the overhead of that sample.

Tests with best and worst case did give 'middle' and 'median' a ~1.5x advantage over random, which makes sense for sorted and reverse-sorted data where the middle and median from three indices including the middle will give you the true median.

I settled on 'median' as the best pivot selection method, as it was within 5% of 'middle' as the best performer for all tests but has more flexibility than 'middle'.

3) Mergesort vs Quicksort Experiments

Quicksort is faster but the same order of magnitude for averageCase, and an order of magnitude faster for best and worst case.



4) Do the actual running times of your sorting methods exhibit the growth rates you expected to see?

They seem to grow at an O(nlgn) rate from inspection, and charting n vs t/nlgn for Best Merge gives something close to a constant value. Close-ish.